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| **Fiscal Year:** | FY 2022 | **Task Last Updated:** | FY 07/01/2022 |
| **PI Name:** | Sundaresan, Alamelu Ph.D. | | |
| **Project Title:** | Antibody Somatic Hypermutation Under Spaceflight Conditions | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology | | |
| **Space Biology Cross-Element Discipline:** | (1) Immunology | | |
| **Space Biology Special Category:** | (1) Cell Culture | | |
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| **City:** | Houston | **State:** | TX |
| **Zip Code:** | 77004 | **Congressional District:** | 18 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2014 ILSRA--Flight Opportunities for Space Life Sciences (non-US proposers) |
| **Start Date:** | 09/01/2020 | **End Date:** | 08/31/2023 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | 2 |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Griko, Yuri | **Contact Phone:** | 650-604-0519 |
| **Contact Email:** | Yuri.V.Griko@nasa.gov | | |
| **Flight Program:** | ISS | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | None. | | |
| **COI Name (Institution):** | Frippiat, Jean-Pol Ph.D. ( European Space Agency (ESA) PI: Lorraine University, France ) | | |
| **Grant/Contract No.:** | 80NSSC20K1308 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Dr Alamelu Sundaresan is the U.S. Co-Investigator on this European Space Agency (ESA)-sponsored project. ESA/CNES (Centre National d'Etudes Spatiales/ French National Centre for Space Studies) Principal Investigator is Jean-Pol Frippiat, Ph.D., EA7300, Lorraine-University, France.<br>Immunity is a key biological system affected by spaceflight. Our previous space experiment suggested that spaceflight conditions decrease the frequency of antibody somatic hypermutations (Bascove et al., 2011, FASEB Journal). This observation could imply less efficient maturation of antibody affinity in space, which would in turn decrease the efficiency of the immune response.<br><br>Because these data were obtained using a small number of amphibians (n=2), we would like to 1) confirm these results using a constitutively hypermutating human B cell line and 2) understand how the genetic machinery that introduces these mutations is altered in space.<br><br>To reach these goals, we plan to culture a constitutively hypermutating human B cell line in the International Space Station (ISS). Cells will be grown onboard the ISS and samples of this culture will be fixed in-flight to determine spaceflight-induced changes. The same experiment will be performed on Earth as a control. Space samples will be returned, when possible, to laboratories where they and control samples will be used for 1) somatic hypermutation analyses to determine if their frequency and pattern along an immunoglobulin VDJ (variability, diversity, and joining) rearrangement are modified and 2) gene expression profiling using microarrays to understand how the somatic hypermutation machinery that introduces these mutations is affected by spaceflight conditions. These analyses will expand our knowledge regarding the effects of spaceflight conditions on antibody somatic hypermutations and consequently, contribute to a better understanding of spaceflight-associated weakening of the immune system, which is an important issue for the preparation of long-term crewed space missions. Furthermore, because several studies have suggested that age-related deficiencies in the generation of high-affinity antibodies might be due to impaired somatic hypermutation, this project could also provide useful data for understanding age-related changes in the immune system. The ISS flight experiment will be manifested in future and the experiment time line is thus tentative and could be altered.<br><br>Bascove M, Guéguinou N, Schaerlinger B, Gauquelin-Koch G, Frippiat JP. Decrease in antibody somatic hypermutation frequency under extreme, extended spaceflight conditions. FASEB J. 2011 Sep;25(9):2947-55. https:// . PubMed PMID: 21593434. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | These analyses will improve our knowledge about the effects of spaceflight conditions on antibody somatic hypermutations and consequently contribute to a better understanding of spaceflight-associated immune system weakening, an important issue for the preparation of long-term manned space missions. Furthermore, since several studies suggested that age-related deficiency in generating high-affinity antibodies might be due to impaired somatic hypermutations, this project could also provide useful data to understand age-related changes of the immune system. |
| **Task Progress:** | 1. We are acquiring state of the art instrumentation in the lab from Higher Education Emergency Relief Fund (HEERF) funding to the University and Department. We have purchased and are awaiting an Olympus Confocal microscope system, Amnis Flow cytometer, Quant Studio 7.0 PCR, Luminex, Yuri RPM (microgravity, Moon and Mars gravity analog), Cytation 5, cellular imaging and immunoassay apparatus, Nanolive real-time non-photobleaching microscope. Our lab is being built back and reopened in March 2022. We are conducting some possible experiments in our collaborator's lab at University of Texas (UT) McGovern Medical School, where the Principal Investigator (PI) is Adjunct Faculty. 2. Other issues we have had are COVID pandemic lab restrictions and vendor material availability. We have also had equipment loss and are in the process of replacing equipment such as a -80 degrees C freezer, cell culture incubator, etc. 3. We plan to expand signaling experiments once our lab is fully operational again. 4. We are preparing to characterize the protein profile of constitutively hypermutating B cells (Ramos) in 1g and the microgravity cell culture analog. We will use a Luminex multianalyte method as previously published (Sundaresan et al 2019). 5. We have optimized the cell growth parameters for the Ramos cell line and frozen cells back. This was all lost due to equipment malfunction. Now a re-do is slated for end of July 2022. 6. We have also performed some basic fluorescent techniques on these cells in preparation. Repeat sets are ongoing in June 2022. 7. We have submitted an invited chapter for NASA (International Journal of Molecular Sciences) in August 2021 on the effects of space on the cardiovascular (CV) system and will cite this project as it pertains to B cell hypermutation, which also can affect the performance of the CV system. It is accepted and in press. 8. We also finished working on a manuscript on the effects of space on bone as an invited article for Springer (Editor - Dr. Christian Otto, Universities Space Research Association). In press, July 2022. 9. Contact was made with Dr. Jean-Pol Frippiat, ESA/CNES (Centre National d'Etudes Spatiales/ French National Centre for Space Studies). He has stated that hardware discussions for spaceflight are intermittently progressing with the ESA program Scientist, Dr. Jason Hatton. Updates will be provided as received.<br>Reference: Mann V, Grimm D, Corydon TJ, Krüger M, Wehland M, Riwaldt S, Sahana J, Kopp S, Bauer J, Reseland JE, Infanger M, Mari Lian A, Okoro E, Sundaresan A. Changes in Human Foetal Osteoblasts Exposed to the Random Positioning Machine and Bone Construct Tissue Engineering. Int J Mol Sci. 2019 Mar 18;20(6):1357. <a target="_blank" href="https://doi.org/10.3390/ijms20061357">https://</a> ; <a target="_blank" href="http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=pubmed&dopt=Abstract&list_uids=30889841">PMID: 30889841</a>; PMCID: PMC6471706. |
| **Bibliography Type:** | Description: (Last Updated: 07/02/2026) |
| **Books/Book Chapters** | Sundaresan A, Mann V, Okoro E. "Effects of space flight on bone and skeletal tissue." in "Handbook of Space Pharmaceuticals." Ed. Y.V. Pathak, M. Araújo dos Santos, L. Zea L. Cham, Switzerland: Springer International Publishing, 2022. p. 323-330. https://doi.org/10.1007/978-3-030-05526-4_21 , Apr-2022 |